

Comprehensive Road Scene Understanding for Autonomous Driving

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Abstract

This report presents our work on comprehensive road scene understanding for autonomous driving, covering semantic, instance, panoptic segmentation, and anomaly segmentation for unknown object detection. We compare ERFNet and EoMT on semantic segmentation, fine-tune EoMT on Cityscapes, and evaluate several post-hoc anomaly detection methods (MSP, Max Logit, Max Entropy, and RbA) on multiple benchmarks. You can find our github here: <https://github.com/GermanRyjluk/FAIMDL-project>

1. Introduction

Robust scene understanding is a critical requirement for safe autonomous driving systems. Modern deep networks achieve high accuracy in closed-set environments, but they happen to fail when they encounter unknown or out-of-distribution (OoD) objects not seen during training [6]. This limitation is particularly dangerous in real-world driving, where unexpected obstacles such as animals or debris can appear on the road at any time and create software failure.

Segmentation architectures. We studied two families of models. ERFNet [13] is a lightweight CNN for real-time semantic segmentation, based on factorized residual blocks. We used it as our pixel-based baseline. The mask architecture family, introduced by MaskFormer [3] and refined by Mask2Former [4], reformulates segmentation as set prediction using learned queries. EoMT [8] further simplifies this by leveraging a DINOv2 [12] backbone. This model was pre-trained and self-supervised on 142M images, showing that a strong visual encoder alone suffices for high-quality segmentation. We evaluated these models on the Cityscapes [5] benchmark for semantic segmentation and the COCO [10] benchmark for panoptic segmentation. The panoptic segmentation task itself was formally defined by Kirillov *et al.* [9].

Anomaly segmentation. Detecting OoD objects is critical for safety. We evaluated on SegmentMeIfYouCan [2] and Fishyscapes [1] benchmarks, they provide standard-

ized evaluation protocols for anomaly segmentation in driving scenes. We apply post-hoc methods including MSP, MaxLogit, MaxEntropy, RbA [11], and temperature scaling [6]. LoRA [7] is discussed as a potential fine-tuning technique.

This project addresses two complementary challenges: (1) understanding and comparing segmentation architectures on standard benchmarks, and (2) evaluating anomaly segmentation baselines for OoD detection in road scenes. We progress from pixel-based (ERFNet) to mask-based (EoMT) architectures, and from standard segmentation to open-world anomaly detection.

The remainder of this report is organized as follows: Section 2 describes our methods, Section 3 presents experimental results, Section 4 discusses findings and limitations, and Section 5 concludes.

2. Methodology

2.1. Architectures

ERFNet. ERFNet [13] is a lightweight CNN for real-time semantic segmentation. It uses factorized residual blocks that decompose 2D convolutions into 1D operations. Given an input image, it produces per-pixel class logits through an encoder-decoder pipeline. While efficient, it is limited to closed-set semantic prediction with no instance-level reasoning.

Mask Architectures. MaskFormer [3] reformulates segmentation as set prediction: N learned queries predict binary masks paired with class scores. Mask2Former [4] improves this with masked attention, forcing each query to attend only within its predicted mask region.

EoMT. EoMT [8] builds on a DINOv2 [12] ViT-B/16 backbone pre-trained self-supervised on 142M images. It appends N learnable queries to the patch tokens in the last transformer blocks, producing mask logits $\mathbf{M} \in \mathbb{R}^{N \times H \times W}$ and class logits $\mathbf{C} \in \mathbb{R}^{N \times (K+1)}$. The per-pixel semantic prediction is:

$$\mathbf{P} = \sum_{q=1}^N \sigma(\mathbf{M}_q) \cdot \text{softmax}(\mathbf{C}_q)_{1:K} \quad (1)$$

where σ denotes sigmoid and the last class (no-object) is discarded.

2.2. Datasets

We used the Cityscapes dataset [5] (2975 train / 500 val images, 19 semantic classes) for segmentation evaluation and fine-tuning. For anomaly segmentation, we evaluated on five benchmarks from the SegmentMeIfYouCan [2] and Fishyscapes [1] benchmarks: RoadAnomaly21, RoadObstacle21, FS Lost&Found, FS Static, and RoadAnomaly.

2.3. Cross-Dataset Evaluation Strategy

Comparing EoMT trained on COCO [10] (133 classes) against Cityscapes ground truth (19 classes) requires a class remapping strategy. We define a mapping from COCO continuous IDs to Cityscapes trainIDs based on semantic correspondence (e.g., COCO *car* $\rightarrow$ CS *car*). Classes without a clear equivalent (e.g., COCO *pizza*) are assigned ignore index 255 and excluded from the mIoU computation. Both models are evaluated using the same confusion matrix-based mIoU pipeline to ensure fairness.

The evaluation metric is the mean Intersection over Union (mIoU), computed from the confusion matrix $\mathbf{M} \in \mathbb{R}^{K \times K}$:

$$\text{IoU}_c = \frac{M_{cc}}{\sum_j M_{cj} + \sum_i M_{ic} - M_{cc}} \quad (2)$$

$$\text{mIoU} = \frac{1}{K} \sum_{c=1}^K \text{IoU}_c \quad (3)$$

where M_{cc} corresponds to true positives, $\sum_j M_{cj} - M_{cc}$ to false negatives, and $\sum_i M_{ic} - M_{cc}$ to false positives for class c .

2.4. Fine-Tuning Strategy

To adapt EoMT-COCO to the Cityscapes domain, we replace the 134-class prediction head with a new 20-class head (19 Cityscapes classes + no-object) initialized randomly. The DINOv2 backbone is frozen; only the class head, mask head, and query embeddings are trained (7.1% of total parameters). We use AdamW (lr = 10^{-4} , weight decay 0.05) with automatic mixed precision (AMP) for 5 epochs on the Cityscapes training set (2975 images, batch size 2, input resolution 640×640).

2.5. Anomaly Segmentation Methods

Post-hoc anomaly segmentation methods derive an anomaly score from the model’s output without retraining. We evaluated three pixel-based methods on ERFNet and four mask-based methods on EoMT:

MSP [6] assigns a high anomaly score to pixels where the maximum softmax probability is low: $s = 1 - \max_c \text{softmax}(\mathbf{z})_c$.

MaxLogit [6] uses the maximum raw logit as confidence: $s = -\max_c z_c$.

MaxEntropy is used to measure the uncertainty of the previsions via the entropy of the softmax distribution: $s = -\sum_c p_c \log p_c$.

RbA [11] is specific to mask architectures. It computes an anomaly score which relies on the degree to which a pixel is rejected by all queries, leveraging the explicit mask predictions of EoMT.

Panoptic segmentation [9] brings together semantic and instance segmentation into a single unified framework, it handles *stuff* classes (like sky or road) at the semantic level, while treating *things* classes (like cars or people) as individual instances.

Temperature Scaling calibrates model confidence by dividing logits by a scalar T before softmax, as described in Section 3.

3. Experimental Results

3.1. Qualitative Comparison (Step 4)

Fig. 1 shows predictions of both EoMT variants on three Cityscapes validation images. EoMT-CS produces spatially coherent semantic maps closely aligned with the ground truth, correctly segmenting road, sky, vegetation, and individual vehicles. EoMT-COCO, visualized as a panoptic prediction, segments individual instances with distinct colors — each vehicle and pedestrian receives a unique identity. However, it struggles with road-specific classes absent from COCO (e.g., *sidewalk*, *pole*, *terrain*), which appear as unclassified regions. This qualitative difference directly reflects the class space mismatch: EoMT-COCO was never trained to recognize these driving-specific categories.

The panoptic output of EoMT-COCO also highlights a fundamental difference in task formulation: while EoMT-CS assigns a single label per pixel based on category, EoMT-COCO distinguishes individual instances — two adjacent cars receive different colors even though they belong to the same semantic class. This instance-level awareness is the key advantage of panoptic segmentation over pure semantic approaches.

3.2. Quantitative Comparison (Step 4)

Tab. 1 reports per-class IoU on Cityscapes val for all three models. EoMT-CS achieves 79.5% mIoU but EoMT-COCO reaches only 23.1%, with strong performance on shared classes (*person*: 68.7%, *car*: 63.6%, *bicycle*: 68.3%) but 0% on road-specific classes that are absent from COCO (*road*, *sidewalk*, *sky*).

3.3. Fine-Tuning Results (Step 5)

The fine-tuning of the prediction head only (7.1% of parameters) for 5 epochs raises EoMT-COCO from 23.1% to

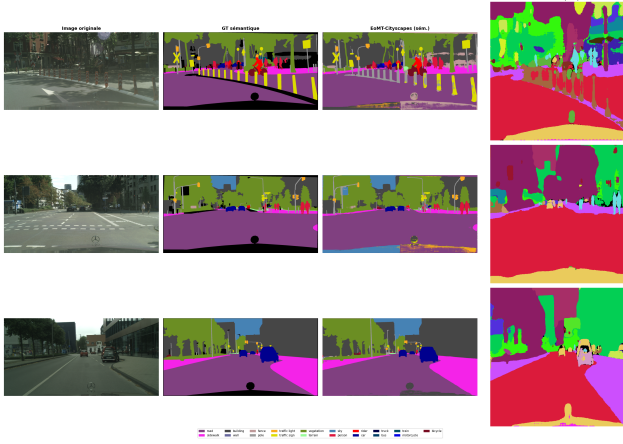


Figure 1. Qualitative comparison. Left to right: input, ground truth, EoMT-CS semantic prediction, EoMT-COCO panoptic prediction.

Table 1. Per-class IoU (%) on Cityscapes val. EoMT-COCO is evaluated after remapping to the 19 Cityscapes classes (unmapped classes = ignore).

Class	EoMT-CS	EoMT-COCO	Fine-tuned
Road	98.2	0.0	97.6
Sidewalk	86.6	0.0	80.2
Building	93.5	11.0	88.9
Wall	63.2	24.9	48.6
Fence	59.4	1.6	46.2
Pole	69.2	0.0	29.5
Traffic light	72.3	63.6	47.9
Traffic sign	77.9	31.3	58.4
Vegetation	92.5	0.1	87.5
Terrain	64.5	0.0	64.0
Sky	94.5	0.0	92.5
Person	83.7	68.7	65.2
Rider	67.2	0.0	28.3
Car	95.0	63.6	92.0
Truck	84.6	10.2	69.6
Bus	88.8	58.5	78.4
Train	72.3	0.0	43.6
Motorcycle	67.4	37.7	34.2
Bicycle	79.4	68.3	61.5
mIoU	79.5	23.1	63.9

63.9% mIoU. From EoMT-coco to fine-tuned some Road-specific classes obtain much better results: *road* jumps from 0% to 97.6%, *sky* from 0% to 92.5%. The remaining gap with EoMT-CS (−15.6 points) is expected since the DI-NOv2 backbone remains frozen and was not adapted to the Cityscapes visual domain. Unfreezing the last ViT blocks could close this gap even further.

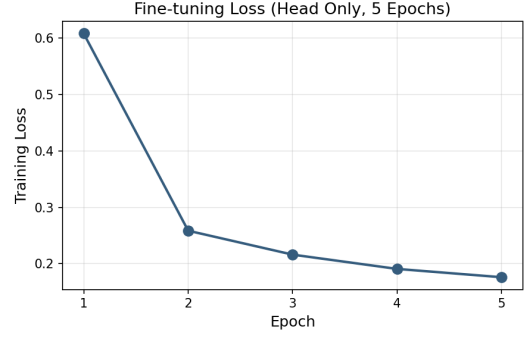


Figure 2. Training loss during fine-tuning of EoMT-COCO on Cityscapes (head only, 5 epochs).

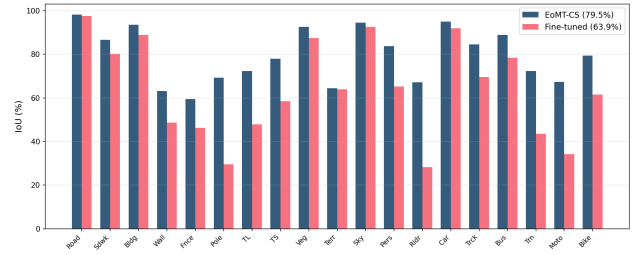


Figure 3. Per-class IoU comparison between EoMT-CS and EoMT-COCO fine-tuned on Cityscapes.

Table 2. ERFNet anomaly segmentation (mIoU = 72.2%). AuPRC / FPR95 (%).

Method	RA-21		RO-21		FS L&F		FS Static		Road An.	
	AP	FPR	AP	FPR	AP	FPR	AP	FPR	AP	FPR
MSP	24.0	72.3	0.67	89.7	1.30	79.9	3.49	63.3	10.8	88.5
MaxLogit	31.9	72.4	1.18	80.3	2.55	72.2	4.24	65.9	13.3	82.5
MaxEntropy	26.1	72.6	0.89	89.6	1.92	79.6	4.42	63.2	11.1	88.6

3.4. Anomaly Segmentation Results (Steps 7–8)

Tab. 2 is the report for the ERFNet post-hoc anomaly baselines and Tab. 3 presents the results of EoMT across all three checkpoints.

3.5. Temperature Scaling (Step 8)

The Temperature scaling methods divides the logits by a scalar T before applying softmax, here is the new equation for model calibration:

$$\text{MSP}(T) = 1 - \max_c \text{softmax}\left(\frac{\mathbf{z}}{T}\right)_c \quad (4)$$

We evaluate $T \in \{0.5, 0.75, 1.1\}$ on RoadAnomaly21. Results in Tab. 4 show that temperature scaling has minimal impact on anomaly detection performance for all three EoMT checkpoints, with AuPRC variations generally below 1.5 points. One exception is EoMT-COCO at $t =$

Table 3. EoMT anomaly segmentation. AuPRC / FPR95 (%). EoMT-CS mIoU=79.5%, EoMT-FT mIoU=63.9%, EoMT-COCO mIoU=23.1%.

Model	mIoU	Method	RA-21		RO-21		FS L&F		FS Static		Road An.	
			AP	FPR	AP	FPR	AP	FPR	AP	FPR	AP	FPR
EoMT-CS	79.5	MSP	36.7	75.8	30.3	93.7	7.6	80.1	22.1	94.1	31.5	75.6
		MaxLogit	20.1	75.6	2.2	94.3	0.3	79.3	1.3	94.2	12.7	75.4
		MaxEntropy	47.8	75.8	42.7	95.4	25.7	79.8	42.2	94.6	47.7	76.1
		RbA	26.8	75.4	20.9	96.7	1.3	69.5	6.7	92.3	17.1	74.0
EoMT-FT	63.9	MSP	23.9	83.0	10.0	79.4	0.2	86.0	1.6	79.2	14.1	81.7
		MaxLogit	25.3	74.8	8.6	79.4	0.3	85.9	2.0	70.2	14.6	80.7
		MaxEntropy	25.0	80.4	11.5	73.0	0.2	85.9	1.9	75.4	15.3	80.8
		RbA	18.4	80.5	19.0	89.8	0.3	95.6	1.5	91.6	14.8	85.5
EoMT-COCO	23.1	MSP	21.4	77.9	8.6	90.5	0.3	93.9	1.5	84.0	9.5	94.9
		MaxLogit	21.3	77.8	9.8	90.6	0.3	96.2	1.5	85.7	9.5	95.1
		MaxEntropy	22.1	77.9	11.0	90.5	0.3	94.3	1.5	84.0	9.6	94.9
		RbA	25.2	68.6	4.9	90.5	0.2	82.1	1.4	95.2	13.2	80.4

Table 4. Temperature scaling on RoadAnomaly21. AuPRC / FPR95 (%).

Method	EoMT-FT		EoMT-COCO		EoMT-CS	
	AP	FPR	AP	FPR	AP	FPR
MSP	23.9	83.0	21.4	77.9	36.8	75.8
MSP ($t = 0.5$)	22.5	85.4	21.1	100.0	34.2	75.8
MSP ($t = 0.75$)	23.4	84.1	21.4	78.2	35.7	75.8
MSP ($t = 1.1$)	24.1	82.5	21.4	77.9	37.2	75.8
Best t	1.1	—	1.1	—	1.1	—

0.5, which degrades FPR95 to 100%, suggesting that over-sharpening the distribution collapses the score range and loses discriminability entirely.

4. Discussion

4.1. Result Analysis

EoMT-CS vs EoMT-COCO. The 56.4-point gap on Cityscapes between EoMT-CS (79.5%) and EoMT-COCO (23.1%) highlights the cost of domain mismatch. EoMT-COCO performs well on universal object classes shared with COCO (*person*: 68.7%, *car*: 63.6%, *bicycle*: 68.3%). But the model tends to fail on road-specific classes (*road*: 0%, *sidewalk*: 0%, *sky*: 0%) because they are either absent or semantically different in the COCO label space. This confirms that a model trained on a general-purpose dataset cannot directly transfer to a domain-specific task without adaptation.

Impact of Fine-Tuning. When we train the model with

only the prediction head (7.1% of parameters) for 5 epochs, we recover 40.8 mIoU points, with an upgrade of EoMT-COCO from 23.1% to 63.9%. This change benefits the most to Road-specific classes: *road* jumps from 0% to 97.6%, *sky* from 0% to 92.5%. This demonstrates that the DINOv2 backbone already encodes rich visual features transferable to Cityscapes and that only the task-specific heads needed retraining. The remaining 15.6-point gap with EoMT-CS (79.5%) is expected since the backbone was frozen and could not adapt its internal representations to the Cityscapes visual domain.

Fine-grained Analysis. The classes with small or thin structures as *pole* (29.5%) or as *rider* (28.3%) remain challenging for the fine-tuned model. This is likely due to the 640×640 input resolution inherited from COCO training that is lower than the 1024×1024 used by EoMT-CS. We can conclude that finer structures require higher resolution to be accurately segmented.

4.2. Methodological Limitations

Class Remapping. Our COCO-to-Cityscapes mapping is necessarily approximate because some Cityscapes classes have no direct COCO equivalent (*pole*, *rider*), and some mappings also involve semantic approximations (e.g., COCO *stop sign* is mapped to Cityscapes *traffic sign*). The mIoU of EoMT-COCO is computed only on mapped classes and it may overestimate performance on the subset of shared classes, while ignoring the model’s inability to predict unmapped ones.

Fine-Tuning Scope. We only fine-tuned the prediction head (Phase 1). Gradually unfreezing the last ViT blocks

(Phase 2) or applying LoRA [7] could contribute to close the gap with EoMT-CS, but those methods were not explored due to Colab time constraints.

Input Resolution. The fine-tuned model uses 640×640 input resolution from COCO, while EoMT-CS uses 1024×1024 . This resolution mismatch disadvantages the fine-tuned model on small objects and could partially explain the performance gap.

4.3. Anomaly Segmentation Analysis

ERFNet vs EoMT. Despite its lower mIoU (72.2% vs 79.5%), ERFNet achieves competitive anomaly detection against EoMT on most methods and datasets (e.g., MaxLogit AuPRC 31.9% vs 20.1% for EoMT-CS on RA-21). However, EoMT-CS MaxEntropy outperforms all ERFNet baselines on RA-21 (47.8% vs 26.1%), suggesting that entropy-based scoring benefits more from the richer mask-based representation. The relative strength of ERFNet on MSP and MaxLogit can be explained by the nature of the confidence scores: the per-pixel softmax of ERFNet produces a direct probability distribution over 19 classes, yielding high anomaly scores when confidence is spread across all classes. EoMT combines mask and class logits via an einsum operation, which may produce well-distributed scores even for anomalous regions, making MSP and MaxLogit less discriminative for threshold-based detection.

RbA vs other methods. RbA [11] does not consistently outperform MSP or MaxEntropy on EoMT in our experiments. This is surprising given that RbA was specifically designed for mask architectures. A possible explanation is that RbA requires careful tuning of its rejection threshold, which we apply post-hoc without any dataset-specific calibration. With proper threshold tuning on a validation set, RbA could potentially outperform simpler baselines.

Fine-tuned model. On RA-21, EoMT-CS achieves the highest MSP AuPRC (36.7%), followed by EoMT-FT (23.9%) and EoMT-COCO (21.4%). This ranking is consistent across most methods, suggesting that training directly on Cityscapes produces the most calibrated confidence scores for road scenes. EoMT-FT nonetheless outperforms EoMT-COCO on most datasets and methods, confirming that even partial domain adaptation improves anomaly sensitivity on driving-specific data.

Temperature Scaling. With AuPRC variations below 0.5 points for all temperatures we tested, Temperature scaling has minimal impact across all EoMT checkpoints. This suggests that EoMT’s confidence scores are already well-calibrated relative to the anomaly detection task, and that simple temperature rescaling is insufficient to significantly improve OoD detection for mask-based architectures. More sophisticated calibration methods or architecture-level changes may be needed.

Dataset Analysis. Performance varies significantly across datasets. RoadAnomaly21 yields the highest scores (ERFNet MaxLogit: 31.9%) due to large, clearly visible anomalies. RoadObstacle21 is the most challenging (ERFNet MaxLogit: 1.18%) because obstacles are small and visually similar to road surface. FS Lost&Found and FS Static represent intermediate difficulty levels with different types of anomalies — lost objects vs static foreign objects.

5. Conclusion

We studied and compared pixel-based (ERFNet) and mask-based (EoMT) segmentation architectures for autonomous driving scene understanding. EoMT-CS achieves 79.5% mIoU on Cityscapes, while EoMT-COCO reaches only 23.1% due to domain mismatch, it can be explained because COCO does not include driving-specific classes. Fine-tuning the prediction head alone recovers 40.8 points (63.9% mIoU) which shows that DINOv2’s visual representations transfer to a new domain. Post-hoc anomaly detection baselines show that ERFNet outperforms EoMT on MSP and MaxLogit, while EoMT-CS MaxEntropy achieves the best overall AuPRC on RA-21 (47.8%), suggesting that mask-based representations benefit entropy-based scoring. Temperature scaling has a minimal impact across all checkpoints as we saw Table 4. Future work includes unfreezing deeper ViT layers during fine-tuning and exploring task-specific anomaly training.

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Generative AI was used during this project as a coding-assistant for debugging and for grammar checks.

We relied upon both Claude (Anthropic), and Gemini (Google) to assist us when we needed assistance developing the code. We started by writing the code and core logic ourselves, but if there was a bug that persisted or an error we didn’t understand, we would share our code and context with the AI and ask it to explain the issue or to suggest a fix. This gave us the ability to speed up debugging while retaining complete control of the implementation and design choices.

In regard to this report, all of the text and structure were created by us. The last step before the completion of the report was to have Gemini proofread the document to ensure proper English grammar usage and to correct a few sentences so that the completed document was clear and readable.

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